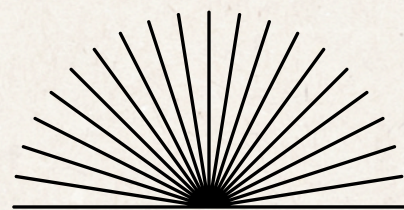


Research Question

- **Topic:** Biological Sex vs. Height in U.S. High School Students.
- **Question:** Is there a significant difference in mean height between males and females?

Method

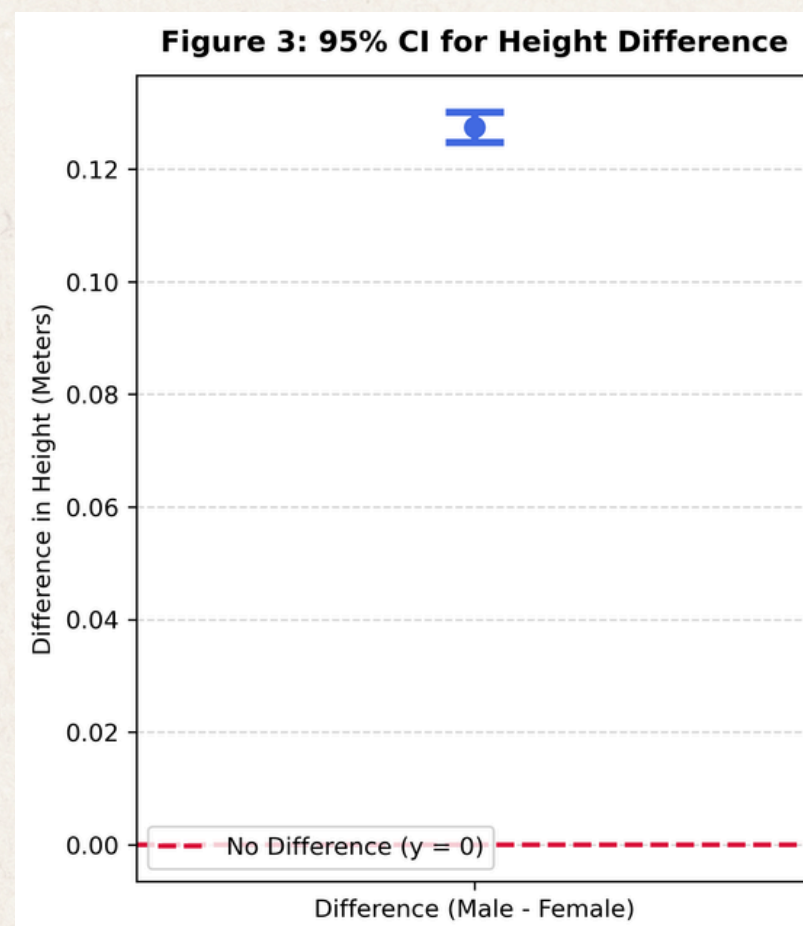
- **Test:** Welch's Two-Sample t-test ($\alpha = 0.05$)
- **Justification:** Accounts for unequal variances; large N satisfies Central Limit Theorem (CLT).



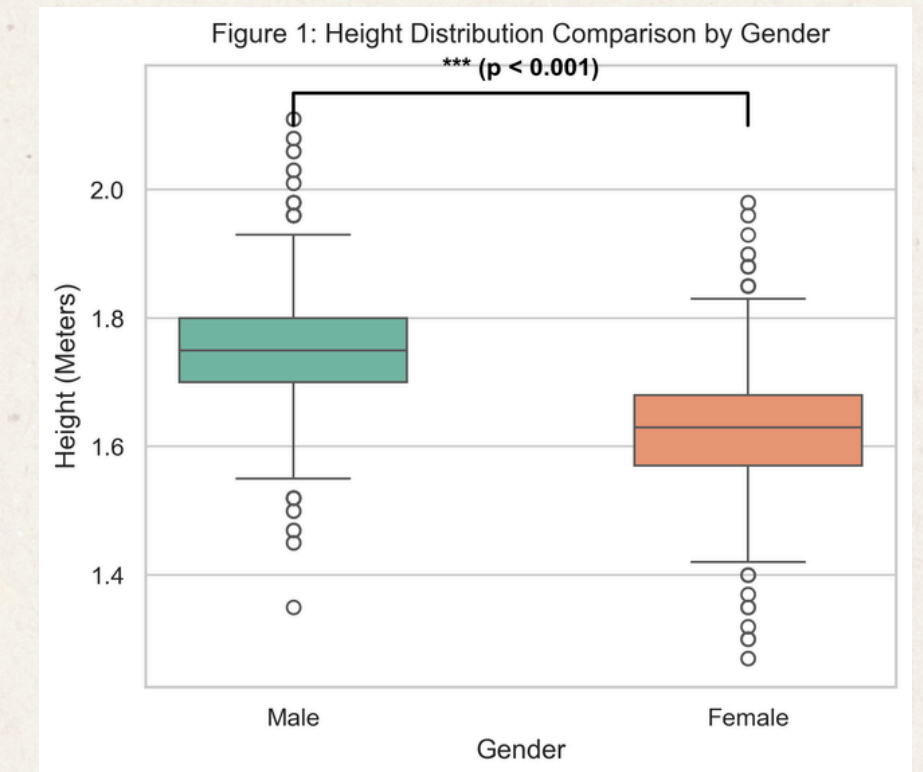
Group Definition

- **Dataset:** CDC 2007 YRBS (N = 13,062)
- **Female (Baseline):** $n_1 = 6,490$
- **Male (Comparison):** $n_2 = 6,572$

Key Results



Response Variable



Conclusion

- **Decision:** Reject H_0
- **Finding:** U.S. male students ($\mu = 1.7574\text{m}$) are significantly taller than females ($\mu = 1.6299\text{m}$) by an average of 12.75cm.